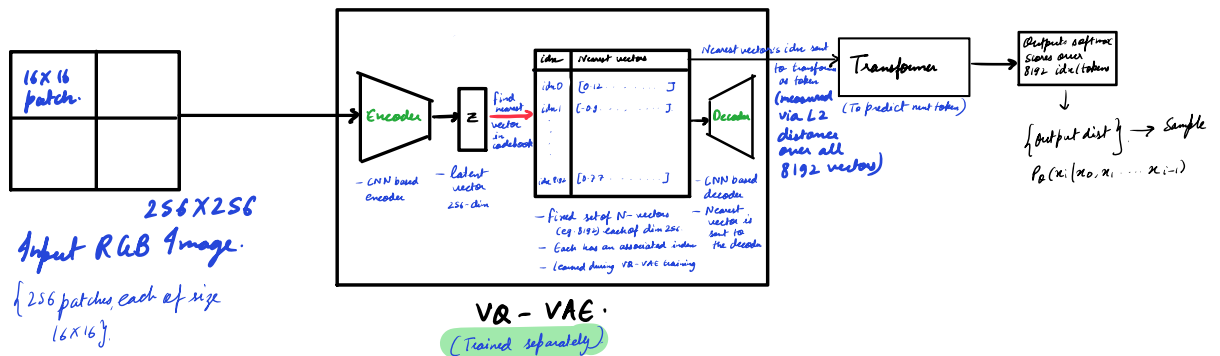


Autoregressive models for Image generation

Instead of generating the entire image at once like the diffusion architecture, an autoregressive model generates the image one token at a time similar to how transformers generate text.

full Pipeline

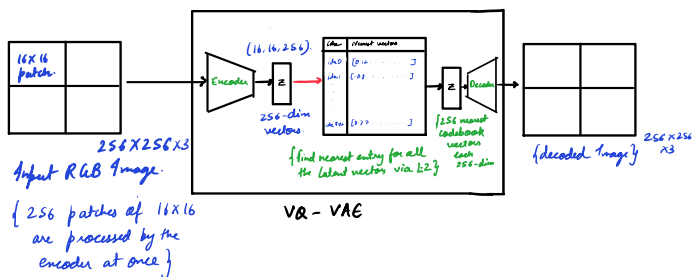


Why VQ-VAE?

Transformers need ^{finite} discrete tokens, not continuous vectors. The codebook creates a ^{finite} vocab. of 8192 tokens for the image and then transformer predicts which of the 8192 idx/tokens come next.

Training

① Phase 1. Train VQ-VAE:



Losses utilized for training:

- Reconstruction Loss:** Comparing reconstructed image with original image.

$$\| \text{original} - \text{image} - \text{reconstructed} \|^2$$
 - Codebook Loss:** Push codebook entries closer to the encoder outputs.

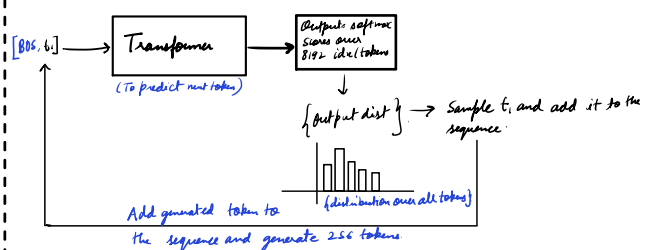
$$\| z - c \|_2^2$$
 {only updates codebook weights}.
 Stop gradient, so that encoder weights don't update from this loss.
 $z \rightarrow$ raw encoder outputs.
 $c \rightarrow$ nearest codebook entry.
 - Commitment Loss:** Push encoder outputs to stay close to their nearest codebook entry.

$$\| z - \text{sg}(c) \|_2^2$$
 {only updates encoder weights}.
- Total Loss = reconstruction + codebook + β * commitment**
 (β is typically 0.25)

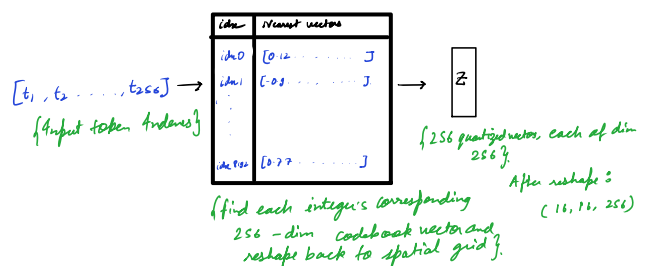
Inference

Inference is performed over 3 steps:

Step 1: Transformer samples 256 tokens autoregressively:

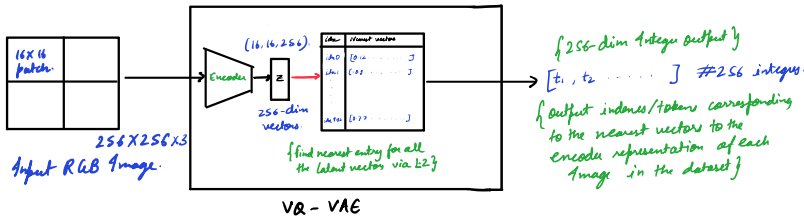


Step 2: Codebook lookup:

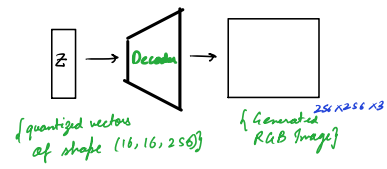


② Phase 2 : Train the Transformer

1. Build the dataset : Run every image through trained VR-VAE (frozen)
& save resulting 256-integer sequences



Step 3: VR-VAE Decoder



2. Train the transformer on the prepared dataset:

